

LAUGHLIN AFB, TX, USA

WMO: 722615

Lat: 29.367N

Lon: 100.783W

Elev: 1082

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 22001

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.5	33.9	9.4	9.2	51.4	13.6	11.4	51.0	28.2	60.0	25.0	55.6	6.5	10	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.9	104.3	72.0	102.1	72.6	100.0	72.7	79.2	89.3	77.7	89.0	76.9	88.5	12.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.9	145.9	83.3	75.1	137.2	81.6	73.9	131.5	81.0	43.6	89.1	42.0	89.1	41.0	88.3	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.0	20.5	18.6	DB	23.7	106.8	4.9	3.3	20.1	109.2	17.3	111.2	14.5	113.0	11.0	115.5
			WB	20.9	80.9	4.2	3.0	17.8	83.0	15.4	84.8	13.0	86.4	9.9	88.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.8	53.3	59.1	65.5	72.8	79.9	85.3	87.3	87.6	81.6	72.6	62.2	54.1
	DBStd	14.04	8.46	9.45	8.38	7.03	6.21	4.28	4.38	3.70	5.39	7.56	7.95	7.90
	HDD50	153	63	27	6	0	0	0	0	0	0	1	9	47
	HDD65	1207	370	204	99	17	2	0	0	0	0	26	142	347
	CDD50	8125	167	282	487	684	926	1060	1155	1164	949	702	375	174
	CDD65	3706	8	40	116	251	463	610	690	699	500	262	58	9
	CDH74	45042	124	421	1028	2521	5257	7992	9539	9656	5671	2331	398	104
	CDH80	24190	19	138	358	1104	2695	4524	5697	5763	2908	897	73	14
(14) Wind	WSAvg	9.4	8.1	8.9	10.1	10.9	10.9	11.3	10.5	9.7	8.2	8.7	8.1	7.6
(15) Precipitation	PrecAvg	20.6	0.7	0.7	1.4	1.4	2.8	2.2	2.2	2.4	3.0	2.1	1.1	0.7
	PrecMax	34.7	2.4	2.8	3.7	5.1	8.6	6.0	6.9	10.3	7.6	6.7	5.2	3.1
	PrecMin	9.7	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.1	0.0	0.1	0.0	0.0
	PrecStd	6.6	0.7	0.7	1.1	1.4	2.2	1.8	2.2	3.0	2.0	2.0	1.2	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.2	90.6	92.6	99.4	104.7	106.4	108.1	105.9	104.4	95.2	85.9	81.4
		MCWB	57.8	60.0	62.6	65.8	68.5	71.3	72.2	72.5	71.8	70.4	64.7	59.7
	2%	DB	77.4	84.0	88.2	94.7	99.8	102.4	104.3	103.0	99.3	91.3	82.0	76.8
		MCWB	57.6	59.1	62.6	66.3	70.7	72.9	72.2	73.1	72.2	70.7	63.9	60.8
	5%	DB	73.0	79.2	84.0	90.3	96.2	99.6	101.9	101.1	96.4	88.3	78.9	72.6
		MCWB	55.7	60.5	63.2	66.6	71.1	73.3	72.7	73.2	72.4	69.7	63.7	59.4
	10%	DB	68.5	74.4	80.0	86.3	92.6	96.9	99.4	99.3	93.2	85.4	75.2	68.3
		MCWB	54.9	58.6	62.8	65.5	70.7	73.1	73.3	73.0	71.8	69.0	62.8	57.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.1	69.3	72.2	76.7	80.2	81.5	82.1	79.1	78.3	77.1	73.2	69.2
		MCDB	72.2	76.2	78.6	81.8	84.4	90.2	94.5	92.8	89.5	86.2	77.8	73.3
	2%	WB	62.8	67.0	69.8	73.4	77.4	79.0	78.4	77.5	76.9	75.3	70.1	66.0
		MCDB	70.4	73.8	77.7	83.7	86.1	89.4	91.0	90.9	88.0	84.3	75.0	70.8
	5%	WB	60.3	65.1	68.1	71.8	75.8	77.6	77.0	76.6	75.9	73.9	68.1	63.0
		MCDB	67.6	71.4	76.1	81.8	86.1	88.5	89.7	90.0	86.3	82.1	73.5	68.1
	10%	WB	57.3	62.8	66.5	70.5	74.6	76.5	76.0	75.7	75.0	72.4	66.1	59.7
		MCDB	65.1	70.3	74.6	80.1	85.5	87.9	88.8	89.2	85.3	80.3	72.7	65.4
(35) Mean Daily Temperature Range	5% DB	MDBR	24.4	24.3	23.5	24.2	22.0	21.5	21.6	21.9	20.8	21.8	21.7	22.9
		MCDBR	31.4	30.3	28.8	28.8	26.8	24.8	25.1	24.4	24.2	24.2	24.6	29.3
		MCWBR	16.3	15.0	13.1	12.7	9.2	7.3	7.1	6.8	7.2	9.7	12.8	16.1
	5% WB	MCDBR	23.6	21.0	21.8	21.1	20.7	20.1	20.6	21.3	18.7	17.9	18.0	21.3
		MCWBR	14.5	13.9	12.3	11.2	9.4	8.1	7.2	6.5	6.3	8.6	12.7	15.2
(40) Clear-Sky Solar Irradiance	taub		0.299	0.304	0.326	0.373	0.391	0.400	0.411	0.392	0.383	0.344	0.319	0.302
	taud		2.557	2.550	2.501	2.384	2.387	2.407	2.413	2.464	2.431	2.538	2.536	2.550
	Ebn at Noon		300	308	305	292	285	281	278	283	283	290	290	294
	Edh at Noon		27	30	33	39	39	38	37	35	35	30	27	26
(44) All-Sky Solar Radiation	RadAvg		1043	1282	1602	1869	1936	2179	2160	2064	1693	1381	1098	947
	RadStd		119	177	145	176	188	164	185	115	145	187	138	122

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	-1.51	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page